Characteristics of cyclical development of a market economy in modern conditions and their reflection in economic theory

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Structure

Structure

Relevance of the research topic

- **The nature and scale of the phenomenon:** the capitalist economy reproduces the conditions of general overproduction crises with regular periodicity, while cyclical dynamics permeate all spheres of economic life, requiring consideration in the development of economic theories and models, as well as in the implementation of state economic policy;
- **Historical variability and contemporary practice:** the historical development of the stages of capitalism and individual components of the capitalist economy do not eliminate the objective foundations of the cycle, but rather modifies its nature and manifestations;
- **Lack of systematization and classification of theories:** the existing systematization of theoretical concepts of the cycle in the scientific literature is incomplete, and the system of criteria for classifying and evaluating theories is methodologically flawed;
- **Misconceptions and ideological bias:** most of the existing cycle theories are predominantly positivist in nature and based on a metaphysical understanding of capitalism, describing the cycle primarily mechanistically through the lens of simple quantitative changes, rather than in the context of the system's dialectical self-development mediated by its immanent contradictions.

Purpose and objectives of the research

Purpose of the research:

- Theoretical and empirical justification of the industrial cycle as a key specific form of the dialectical development of the capitalist economy at its current specific historical stage.

Objectives of the research:

1. Define the concept of a cycle, systematize approaches to its definition and research;
2. Identify the key elements of the cycle theory and systematize theoretical concepts based on them;
3. Examine the mechanism of the industrial cycle from the perspective of dialectical development;
4. Describe the reproduction of fixed capital as a key factor of the industrial cycle;
5. Conduct concrete historical analysis of the crises and the cycle in the US economy;
6. Perform an empirical analysis of indicators and measures of the industrial cycle in the US economy.

Scientific novelty of the research (1)

1. The most comprehensive systematization of various theoretical concepts of the cycle in the economy has been carried out to date, using the author's classification system developed within the framework of this work, based on key aspects of analysis and fundamental debatable issues of the cycle theory;
2. The main and secondary contradictions of the capitalist economy and its constituent elements underlying the industrial cycle have been highlighted and specified, and their exacerbation and resolution during the crisis of overproduction have been described in detail. The role of the industrial cycle as a specific form of the dialectical development of the capitalist economy has been substantiated, where its main contradictions remain unchanged, while secondary contradictions are partially and temporarily resolved through force during the crisis of overproduction, and then reproduced again at a higher stage of development of the capitalist economy;

Scientific novelty of the research (2)

3. Based on the principles of Marxist political economy, the author has developed a model of periodic mass renewal of fixed capital for the case of simple reproduction, vividly illustrating the mechanism of periodization and synchronization of the moments of renewal of fixed capital between various enterprises in the capitalist economy;
4. Using a specific historical and political-economic analysis, all economic crises in the US economy have been systematized and classified based on their dates and typology, and the periods and phases of the industrial cycle in the US economy from 1919 to the present have been identified. The dynamics of the organic composition of capital and the rate of profit have been calculated and analyzed based on current statistical data, in relation to the phases of the industrial cycle, empirically confirming the fundamental principles of Marxist theory regarding the long-term trend of increasing the organic composition of capital and decreasing the rate of profit.

Theoretical and practical significance of the research

Theoretical significance:

- Further development of the Marxist theory of crises, improvement of the classification of economic cycle concepts, research on modern forms of capitalist cyclicity, and a deeper understanding of the historical limits of state regulation of the capitalist economy.

Practical significance:

- A tool for the comprehensive analysis of the processes of reproduction of social capital, application in research and educational work, expert assessment of modern economic processes.

Area of research (1)

The content of the research, as well as the main provisions and conclusions, correspond to the passport of the VAK[1] specialty 5.2.1 "Economic Theory":

- **p. 2** Clarification of the economic content of the cycle category, determination of its structure, internal contradictions, and place in the system of categories of social capital reproduction, as well as differentiation of industrial, business, investment, credit, and financial cycles;
- **p. 4** Development and application of a methodology for comprehensive research of the industrial cycle based on the dialectical and historical-materialistic method, the principle of moving from the abstract to the concrete, and the combination of theoretical, historical, and statistical levels of analysis;
- **p. 5** Study of the historical development of cycle theories, identification of the main stages of their formation, and analysis of changes in the understanding of the causes and mechanisms of economic crises;
- **p. 6** Systematization of scientific schools and research programs explaining the nature of economic cycles, as well as a comparative analysis of their subject, methodological and theoretical foundations;

[1] Higher Attestation Commission

Area of research (2)

p. 7 Concrete historical study of the emergence and evolution of the industrial cycle, the periodization of crises in the capitalist economy, and the comparison of cyclical dynamics at different stages of capitalist development;

p. 9 Study of macroeconomic patterns of cyclical movement of the capitalist economy, including the relationship between the dynamics of production, accumulation, investment, profit, employment, credit and aggregate demand;

p. 11 Application of political-economic approach to the study of the industrial cycle, in which cyclical dynamics are derived from the internal contradictions of production, accumulation, and reproduction of social capital;

p. 16 Development of theoretical approaches to the study of the industrial cycle as a form of economic fluctuations in the capitalist economy, revealing its internal causes, phase structure, mechanism of movement and historically changing forms of manifestation.

Structure

Eclectic cycle definition

The multitude of competing theoretical and methodological approaches to analyzing the cycle in the economy necessitates the formulation of an eclectic definition as the initial abstraction for further theoretical specification of the concept:

Definition

Economic cycle – oobjectified wave–like movement of aggregate economic activity within the framework of patterns of changes in aggregate output and other related indicators of the economic environment (production, employment, prices, profits, investments etc.), described as a sequential change in the phases of its qualitative states and externally observed in the form of stable deviations from a certain "normal" state or a long-term development trend.

Important

This definition is extremely abstract and mainly descriptive, therefore is not fully reliable and requires further specification.

Methodology of cycle theories

All existing cycle theories can be classified into the following categories based on their methodology:

- *marxist theories:* consider the industrial cycle as a specific form of dialectical development of an internally contradictory capitalist economy within a certain historical stage;[2]
- *bourgeois theories:* consider the phenomenon of the cycle through the lens of the absolutization of the role of individual factors within an internally consistent economic system;[3]
- *eclectic theories:* combine incompatible elements of the two previous categories in various proportions or externally reproduce the conceptual framework of one of them while actually borrowing the substantive provisions of the other.[4]

[2] Marx, Engels, Maksakovskiy, Mendelson et al.

[3] Kitchin, Kuznets, Juglar, Kydland & Prescott, Schumpeter et al.

[4] Kondratiev, Luxembourg, Bukharin et al.

Criteria for analysis and systematization

In order to further systematize scientific works and specify their theoretical provisions, a number of key criteria have been identified:

1. Definition
2. Terms
3. Methodology of research
4. Scientific approaches
5. Tools of research
6. Methods of research
7. Indicators
8. Economy
9. Initial sphere
10. Fundamental causes
11. Factors
12. Period
13. Multicyclicity
14. Phases
15. Regulation

Industrial cycle (1)

Finding

A critical analysis of existing theories reveals that the Marxist theory of the industrial cycle provides the most comprehensive and accurate description of cyclical phenomena within a capitalist economy.

Definition

Industrial cycle – objectively determined, historically specific and relatively periodic form of the movement of expanded reproduction of total social capital, expressed in the regular and sequential alternation of phases of crisis, depression, recovery, and growth, arising from the accumulation, exacerbation, temporary violent resolution, and subsequent reproduction of the internal contradictions of the capitalist mode of production.

Industrial cycle (2)

Indicators	*Industrial production volume* – general dynamics and phases; *GDP* – initial point of general overproduction crisis
Economy	Developed capitalist economy with a predominant share of large-scale machine production; national economy; 1825 – the first crisis of general overproduction in Great Britain
Initial sphere	Industrial production
Fundamental causes	Immanent contradictions of the capitalist system
Factors	Massive periodic renewal of fixed capital

Industrial cycle (3)

Period	$\approx$ 10 years, due to the period of physical and moral depreciation of fixed capital
Multicyclicity	The dominant role of the industrial cycle, partial impact on the dynamics of other cycles[5]
Phases	Four-phase model: crisis $\rightarrow$ depression $\rightarrow$ recovery $\rightarrow$ growth
Regulation	Government regulation can accelerate or delay the onset of a crisis, change its nature or duration, but cannot eliminate it

[5] In particular, the agricultural cycle

Structure

Mechanism of dialectical development

The mechanism of the dialectical development of the system includes four successive stages:

1. *Accumulation* of contradictions of the system in a hidden form through gradual changes in the quantitative ratio of its dialectical opposites;
2. *Aggravation* of the accumulated contradictions of the system in an explicit form to a critical level due to a significant change in the quantitative ratio of its dialectical opposites;
3. *Resolution* of contradictions incompatible with the existence of the current system by violent means by breaking the dialectical unity of its opposites during the crisis;
4. *Removal* of contradictions of the previous system in the course of its denial or transition to a qualitatively new level of development through the formation of a new dialectical unity of opposites.

Finding

Crisis – natural turning point, acting as a constitutive phase of the cyclical development of a system, expressed in the rupture of the former form of unity of its opposites and thereby causing its transition to a qualitatively new state within each subsequent turn of the spiral of dialectical development.

Cycle as a form of dialectical development

The fundamental contradictions of all constituent elements of the capitalist economy within its framework can be classified into two main categories:

- *Main contradictions* – directly define the capitalist economy, cannot be resolved within the framework of capitalism and form the basis of more specific secondary contradictions;[6]
- *Secondary contradictions* – lie on the surface, representing a more concrete and derivative form of the main contradictions, partially and temporarily resolving by force during the crisis of the capitalist economy, reproducing at a higher stage of its development.[7]

Finding

The movement of the capitalist economy should be characterized precisely as an **industrial cycle**, since the temporary partial resolution of its *secondary contradictions* during the crisis does not entail the negation of the capitalist system itself as such, while the *main contradictions* serving as the primary source of its movement remain unchanged at all stages of its development.

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[6] In particular, the contradiction between *social character of the productive forces* and *private ownership of the means of production*

[7] In particular, the contradiction between *high production potential of capitalism* and *low consumption capacity of the masses*

Structure

Characteristics of fixed capital

Definition

Fixed capital – special part of constant[8] capital in the form of machinery and equipment that is not fully consumed within a single production cycle, functions through repeated participation in production, retains its material content and is subject to physical and moral deterioration during the gradual transfer of its own value to the value of finished products.

Features of the fixed capital:

- *Periodicity and mass reproduction:* the need for periodic mass advance payments in full due to physical and moral deterioration;
- *Inertia of reproduction:* significant amount of advance payments, long period of turnover, high maintenance and operation costs, difficulty of capital mobility between industries;
- *Cyclical role:* aggravation of immanent contradictions of the capitalist mode of production during the process of renewal, the development of the material and production base forms the prerequisites for a crisis of general overproduction.

[8] In marxist terminology, *constant capital* – equipment and raw materials (transfers value), *variable capital* – labor (creates surplus value). At the same time, equipment makes up the *fixed capital*, and raw materials and labor make up the *working capital*.

Cyclical role of fixed capital

Mechanism of cycle formation:

1. Necessity of renewal of fixed capital under the pressure of physical wear and favorable market conditions at the beginning of the growth phase;
2. Increasing output of means of production, expansion and technological modernization of the production base;
3. Growth of the output of consumer goods, the aggravation of the contradiction between production and consumption, the prerequisites for the crisis of overproduction;
4. Partial and temporary forced resolution of contradictions during the crisis of overproduction, capital depreciation.

Proposition

Technological modernization causes an increase in the organic composition of capital and a decrease in the rate of profit, while a decrease in the rate of profit necessitates technological modernization at the beginning of the next phase of growth.

Finding

The renewal of fixed capital is both a cause and a consequence of the crisis of general overproduction, causing a cyclical form of movement of the capitalist economy.

Reproduction of fixed capital within the enterprise

The process of capitalist reproduction within a single enterprise can be decomposed into a continuous annual turnover of working capital $wc_t = c_t^{wc} + v_t^{wc}$ and a discrete turnover of fixed capital $o = C_*^o = \sum_{n=t}^{t+(l-1)} \overline{c}_n^o$ for $l \in \mathbb{N}$ years:

Equation

$$\begin{cases} D_t^{wc} \to T_t^{wc} \to P_t^{wc}(c_t^{wc} + v_t^{wc}) \xRightarrow[\text{value increment}]{w_t^{wc}=c_t^{wc}+v_t^{wc}+m_t} T_t^{\prime\, wc} \to D_t^{\prime\, wc}, \quad D_t^{\prime\, wc} > D_t^{wc} \\ D_{t_i^{ce}}^o \to T_{t_i^{ce}}^o \to P_{t\in\{t_i^{ce},\dots,t_i^{da}\}}^o \xRightarrow[\text{value transfer}]{w_t^o=\overline{c}_t^0} T_{t_i^{da}}^o \to D_{t_i^{da}}^o, \quad D_{t_i^{ce}}^o = D_{t_i^{da}}^o \end{cases} \tag{1}$$

Example

In the year $t = 1$ the enterprise advanced fixed capital worth $C_1^o = 1000$ units. Subject to uniform annual depreciation of $\overline{c}^o = 100$ units after $l = 10$ years the fixed capital will require full reimbursement in the amount of $C_{11}^o = C_1^o = 1000$ units.

Reproduction of fixed capital within the economy (1)

Let there be a universal set of enterprises in the economy $\mathrm{F}_{\mathbb{U}}$. Let us define $\mathrm{F}_{\mathbb{U}} \supseteq \mathrm{F}_N = \{f_j\}_{j \in \mathrm{J}^N}$ as a set of $N \in \mathbb{N}$ enterprises, which is a representative sample containing the corresponding selection criteria $\mathcal{P}$ for the analysis of cyclical processes in the capitalist economy enterprise $f_j \in \mathrm{F}_N$:

Equation

$$\begin{aligned} \mathrm{F}_{\mathbb{U}} \supseteq \mathrm{F}_N := \{f \in \mathrm{F}_{\mathbb{U}} \mid \mathcal{P}(f)\} \\ |\mathrm{F}_N| = N,\ N \in \mathbb{N} \end{aligned} \implies \begin{aligned} \mathrm{F}_N = \{f_j \mid j \in \mathrm{J}^N\} \\ \mathrm{J}^N = \{1, 2, \dots, N\} \end{aligned} \tag{2}$$

Reproduction of fixed capital within the economy (2)

In this case, the cumulative stochastic fixed capital renewal flow $K^{\mathrm{s}}(t)$ can be represented as:

Equation

$$K^{\mathrm{s}}(t) = K^{\mathrm{d}}(t) + \underbrace{\sum_{n=0}^{\infty} \rho^n x_{t-nL}}_{\mathcal{X}_t^{\rho}} - \underbrace{\mathbf{1}_{\mathrm{T}^{\mathrm{d,ce}}}(t) \cdot (1-\rho) \sum_{n=1}^{\infty} x_{t-nL}}_{R_t} \tag{3}$$

$K^{\mathrm{s}}(t)$	Cumulative stochastic flow
$K^{\mathrm{d}}(t)$	Base determined flow
$\mathcal{X}_t^{\rho}$	Random deviations in growth phase
R_t	Neutralization flow under competitive pressure
ρ	Deviation neutralization parameter, $\rho \in [0; 1]$

Example

Enterprises $f_j \in \mathrm{F}_N$ in periods $t = 1, 2, 3$ respectively advanced fixed capital in physical volumes $K^{\mathrm{s}}(1) = 10,\ K^{\mathrm{s}}(2) = 8,\ K^{\mathrm{s}}(3) = 5$. In this case, with a period of physical depreciation $l^j = 10,\ \forall j \in \mathrm{J}^N$ excluding deviations $\mathcal{X}_t^{\rho}$ and neutralization R_t within simple reproduction after 10 years fixed capital will be advanced again in the amounts of $K^{\mathrm{s}}(11) = 10,\ K^{\mathrm{s}}(12) = 8,\ K^{\mathrm{s}}(13) = 5$ respectively.

Reproduction of fixed capital within the economy (3)

Finding

The reproduction of fixed capital within capitalist economy acquires a **periodic and massive character**, and the moments of its full advance and subsequent compensation are synchronized **at the beginning of the growth phase** under the influence of **market conditions, fluctuations in rate of profit and competition**, whereas deviations arising from the industry rhythm are offset over time under the pressure of the above mentioned factors and do not entail a violation of the periodicity of this the process.

Factors modifying cycle

Factors modifying cycle can be divided into the following main categories:

Cycle catalysts	trade and speculation, commercial credit and bank loans
Structural changes	monopolization of the economy, financial and fictitious capital, government regulation
Spatial boundaries	capital exports, global value chains, unequal trade exchange

Important

Even under the conditions of modern capitalism the core of the cyclical dynamics of the capitalist economy is still precisely the sphere of industrial production.

Structure

Cycle dating and phasing

Identifying the periods and phases of the industrial cycle, the crises of the capitalist economy are necessary to distinguish:

In scale:

- *general overproduction* – whole national economy
- *partial overproduction* – individual industries or sectors of the economy

In nature:

- *cyclical* – caused by immanent contradictions and reproduction of fixed capital
- *intermediate* – other and mostly exogenous factors

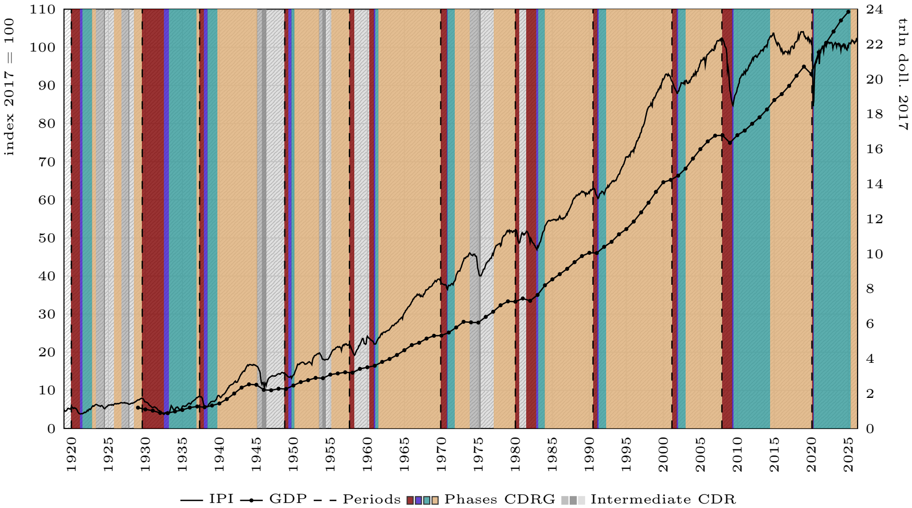
Proposition

Industrial cycle dating and phasing is based on cyclical crises of general overproduction.

Important

Only quantitative data with qualitative analysis of historical conditions comparison makes distinguishing the cyclical crisis of general overproduction from non-cyclical recessions, sectoral imbalances and externally caused economic shocks possible.

Cycle periods and phases (1)



Source: FRED, NBER

Cycle periods and phases (2)

Finding

The standard sequence "crisis → depression → recovery → growth", reflecting the internal logic of the movement of the industrial cycle, may undergo some changes in the real dynamics of the capitalist economy.

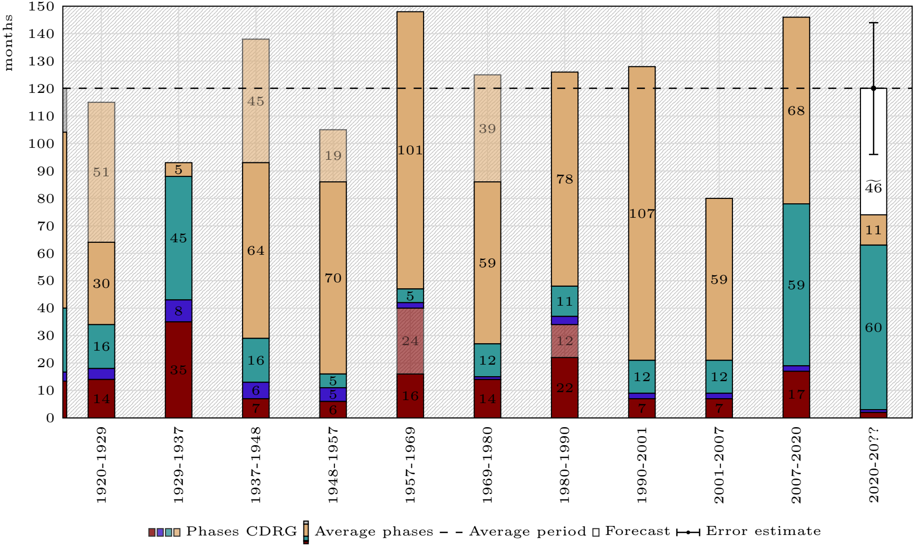
There are two forms of these deviations:

- *intermediate crisis* – occurs under the influence of exogenous factors during the growth phase, after which it can either continue or end[9]
- *intermediate recovery* – occurs during the crisis phase under the influence of a short-term effect of stimulating government policy, after which the crisis phase continues[10]

[9] For example, during the crises of 1973 and 1945 respectively

[10] For example, during the crisis periods of 1957-1961 and 1980-1982

Cycle periods and phases duration (1)



Source: FRED, NBER

Cycle periods and phases duration (2)

Cycle periods and phases duration within the study period was in months:

Duration	Period	Crisis	Depression	Recovery	Growth
Average	120.1	16.6	3.3	23.0	79.5
Adjusted	120.1	13.4	3.3	23.0	64.1
Maximum	148.0	35.0	8.0	60.0	107.0
Minimum	81.0	2.0	1.0	5.0	30.0

Observation

Duration of the depression phase of the industrial cycle decreased from 6-8 months to 1-2 months from 1919 to 2026, which can be attributed to the development of government incentive policies.

Finding

Empirically obtained data confirm the proposition about 10 years average duration of the industrial cycle period, corresponding the period of physical depreciation of fixed capital in the field of industrial production.

Current industrial cycle

The current industrial cycle in the US economy began after the crisis slump in production in 2020 has a number of features:

- An extremely long recovery phase after the 2020 crisis of 60 months with an average duration of 23 months;
- The economy is formally in growth phase relative to the last crisis, according to NBER dating, but in fact the pre-crisis maximum of 2018 of 104 points has not been overcome yet;
- The pre-crisis level of 2007 has hardly been overcome only twice in 2014 and 2018 respectively, and at the moment the index value is still below the maximum of 2007, indicating the stagnation of industrial production.

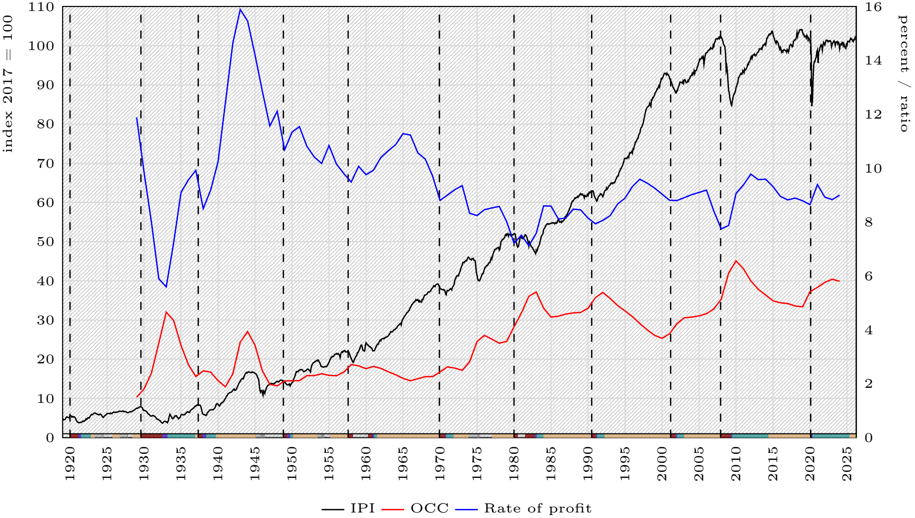
Hypothesis

The next crisis of general overproduction in the United States based on the frequency of the industrial cycle identified in this paper all other things being equal may unfold in the period from 2028 to 2032.

A potential reason for a new cyclical crisis may be the depreciation of capital in the artificial intelligence market due to an inflated financial bubble based on exagerated expectations of future profits.

Structure

Organic composition of capital and rate of profit (1)



Source: FRED, NBER, BEA

Organic composition of capital and rate of profit (2)

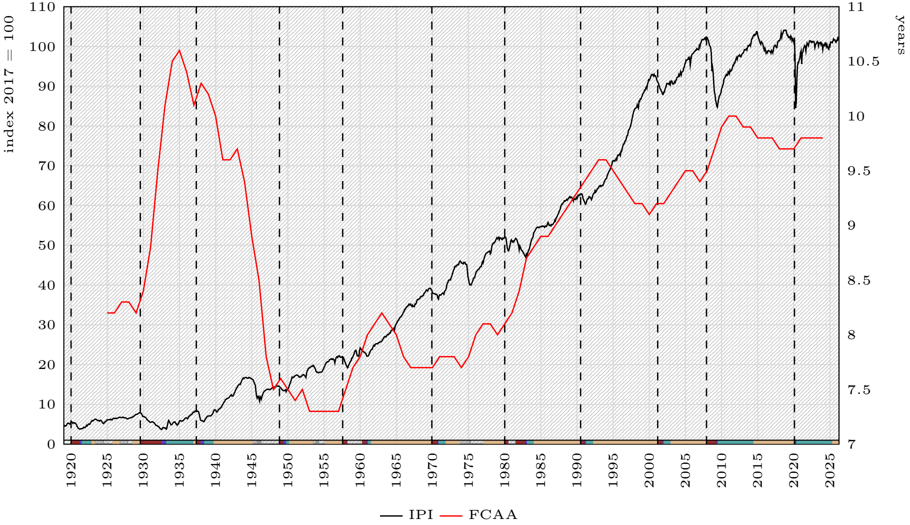
Observation

- **Organic composition of capital** grows at the end of the growth phase due to the commissioning of production facilities created as a result of previous investments; during the crisis its growth is additionally supported by a sharp decline in employment and wages.
- **Rate of profit** moves in the opposite direction, begins to decline from about the middle of the industrial cycle under the pressure of constant capital accumulation outpacing the growth of surplus value, during the crisis it additionally decreases due to the depreciation of capital, after which it temporarily recovers.

Finding

Empirically obtained results on the basis of up-to-date statistical data confirm the fundamental principles of marxist political economy regarding the long-term trend of increasing the organic composition of capital and decreasing the rate of profit.

Fixed capital average age (1)



Source: FRED, NBER, BEA

Fixed capital average age (2)

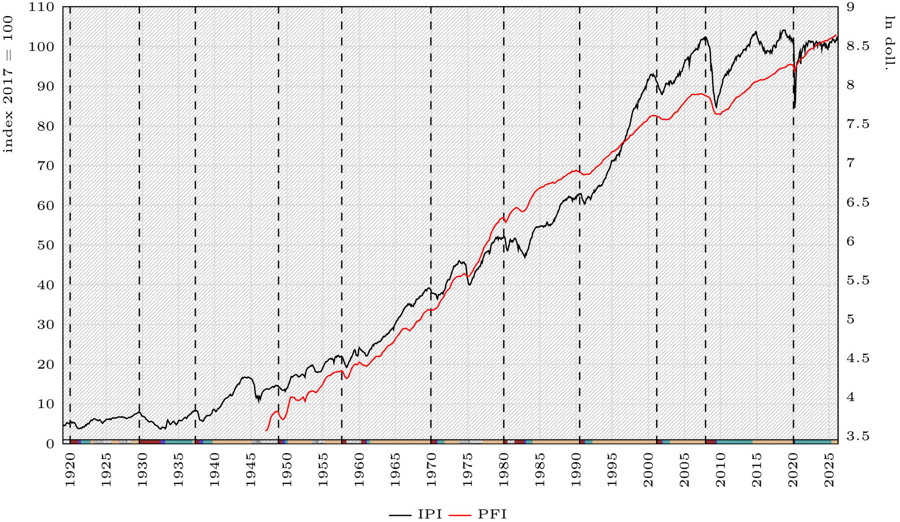
Observation

Average age of fixed capital decreases at the beginning of the growth phase during technological modernization and expansion of production capacities, and increases during the crisis due to the suspension of large investments.

Finding

The dynamics of the average age of fixed capital confirms the propositions about the restoration of production during the recovery phase due to existing production capacities and the massive renewal of fixed assets at the initial stage of the growth phase.

Private fixed investment (1)



Source: FRED, NBER

Private fixed investment (2)

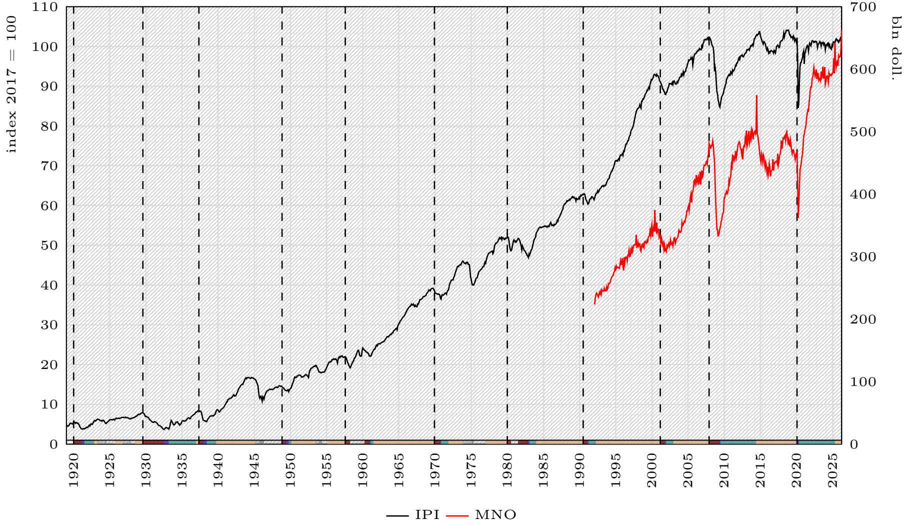
Observation

Private fixed investment generally follows the dynamics of the industrial production index, but at the end of the growth phase begin to decline earlier, and after the recovery phase they recover a little later and more slowly.

Finding

The dynamics of private fixed investment confirms the proposition about the recovery of production due to existing production capacities; a reduction in investment at the end of the growth phase signals the approach of a cyclical crisis of general overproduction.

Manufacturers' new orders (1)



Source: FRED, NBER

Manufacturers' new orders (2)

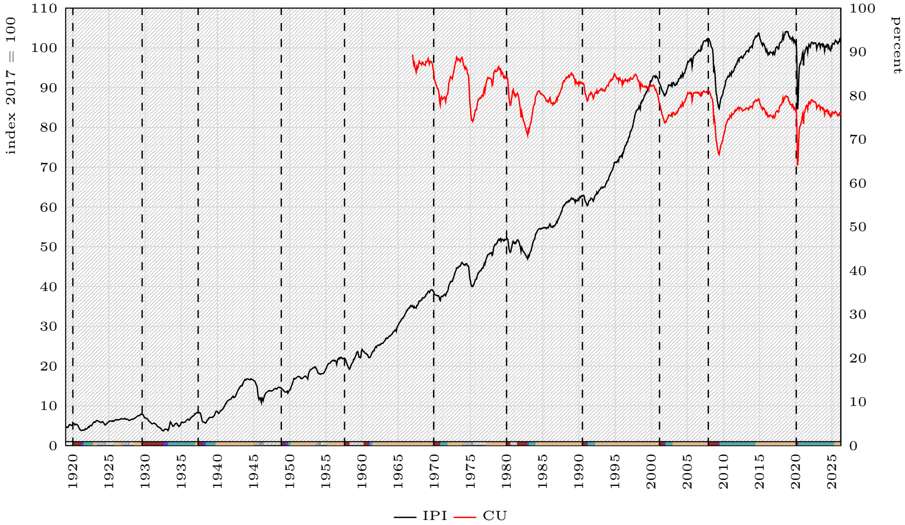
Observation

Manufacturers' new orders generally follow the dynamics of private fixed investment, shrink earlier at the end of the growth phase, and recover with some delay after the recovery phase.

Finding

The slump of the manufacturers' new orders at the end of the growth phase signals the approach of a cyclical crisis of general overproduction.

Capacity utilization (1)



Source: FRED, NBER

Capacity utilization (2)

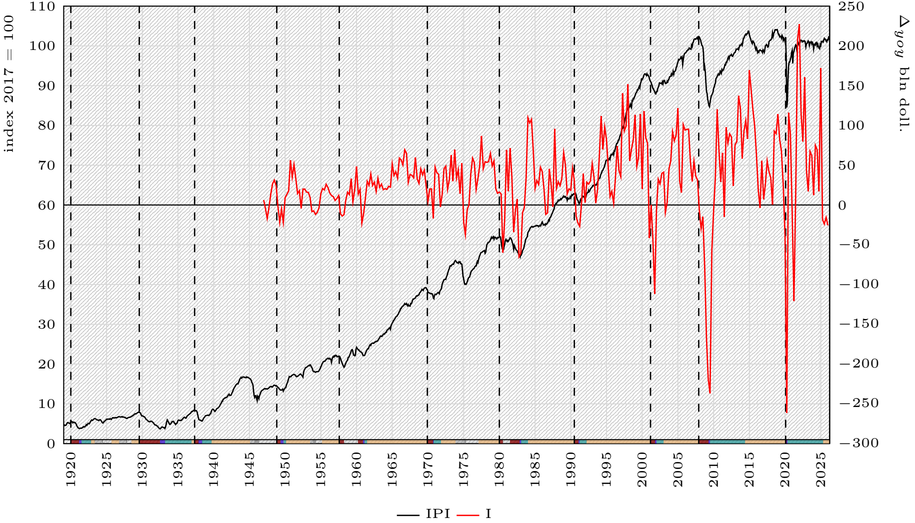
Observation

Capacity utilization exactly repeats the dynamics of the industrial production index and changes synchronously with it, decreases during the crisis, increases during the recovery and growth.

Finding

Capacity utilization can be considered solely as a current indicator of the industrial cycle in order to concretize the assessment of the depth of the decline in production volumes.

Inventories (1)



Source: FRED, NBER

Inventories (2)

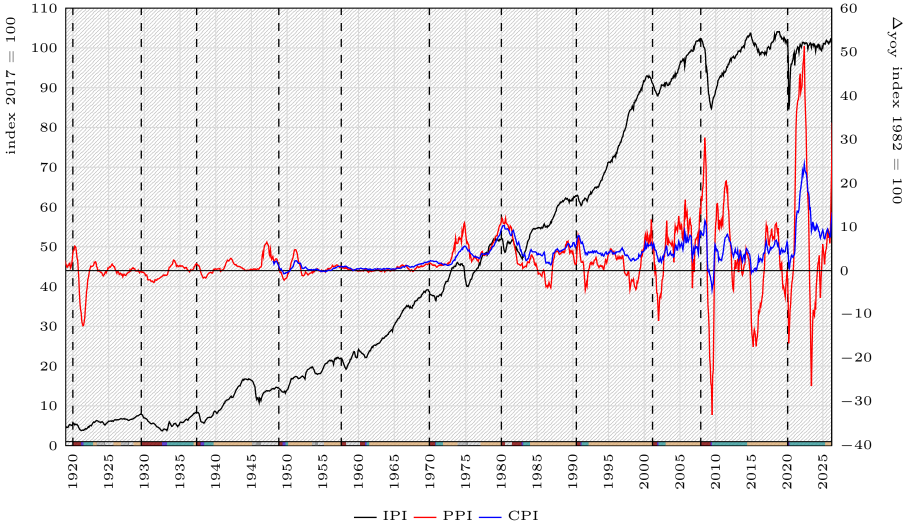
Observation

Inventories decrease during the crisis due to a slump in production and the desire of enterprises to free up associated working capital, then the rate of their accumulation increases until about the middle of the growth phase under the influence of expanding output and expectations of further growth in demand, but slows down towards the end due to the sales difficulties.

Finding

The dynamics of inventories clearly illustrates the aggravation and temporary resolution of the contradiction between the conditions of production and the conditions of realization of surplus value, as well as the process of depreciation of the commodity form of capital during the crisis of overproduction.

Producer and consumer prices (1)



Source: FRED, NBER

Producer and consumer prices (2)

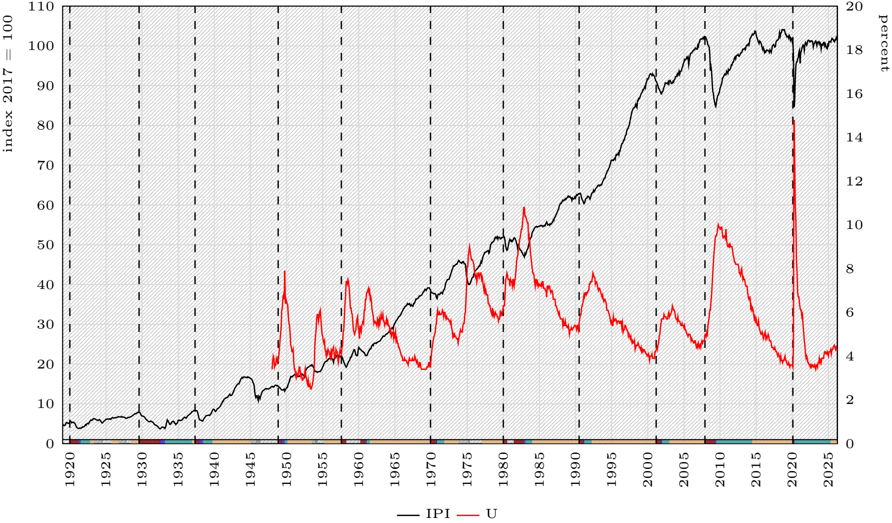
Observation

- The growth rate of the **producer price index** increases at the beginning of the crisis phase under the influence of previously accumulated cost increases and inertia of contract prices, but as the recession deepens it sharply decreases due to reduced demand and difficulties in sales, and then recovers at the beginning of the growth phase.
- The growth rate of the **consumer price index** generally follows the dynamics of the producer price index, but is less volatile and rarely falls below zero under the influence of inflation.

Finding

The dynamics of the growth rates of the producer and consumer price indices clearly illustrate the process of depreciation of the commodity form of capital during the crisis of overproduction.

Unemployment (1)



Source: FRED, NBER

Unemployment (2)

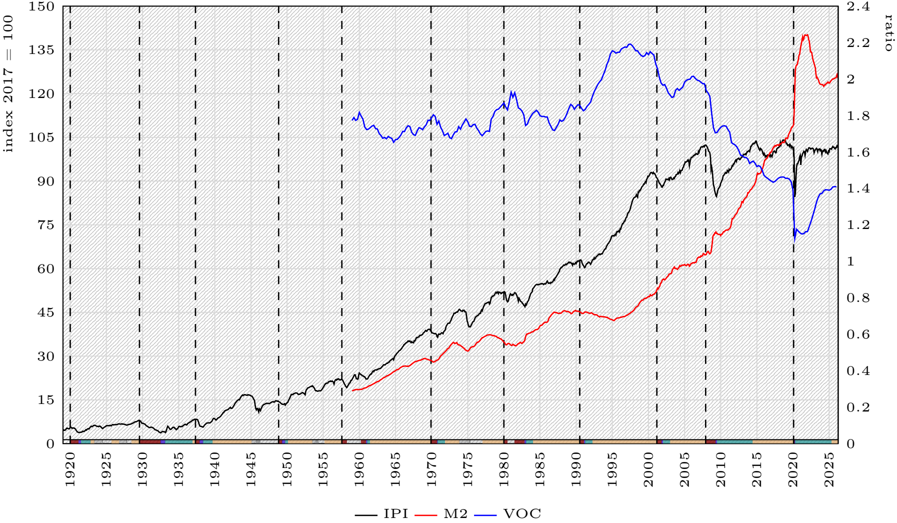
Observation

Unemployment increases sharply at the beginning of the crisis phase under the influence of a slump in production and mass release of labor, starting from the recovery phase employment increases as workers are involved in the production process and wages increase, and by the end of the growth phase the unemployment rate reaches a minimum.

Finding

The dynamics of the unemployment rate clearly illustrates the steady trend of relative overpopulation within the capitalist economy and the replenishment of the reserve army of labor in the context of the crisis of overproduction.

Money supply and velocity of circulation (1)



Source: FRED, NBER

Money supply and velocity of circulation (2)

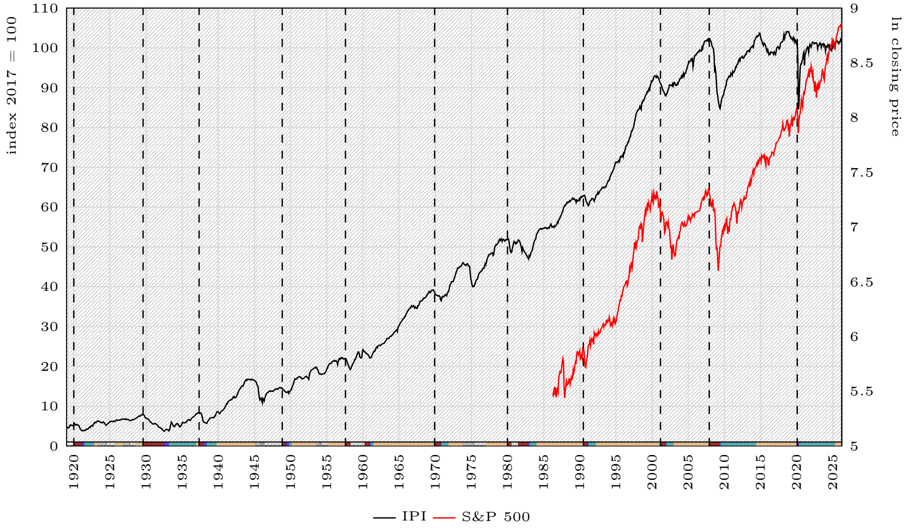
Observation

- **Money supply** during a crisis can either shrink under the influence of a squeeze on bank lending and an raise in non-payments or increase in the case of a stimulating monetary policy.
- **Velocity of circulation** varies inversely with money supply, but always decreases during a crisis under the influence of reduced investment and disruption of payment chains.

Finding

The dynamics of the money supply and velocity of circulation clearly illustrate the process of transformation of the credit system into a monetary one and the growing demand for money as a means of payment in the context of the crisis of overproduction.

Stock market (1)



Source: FRED, NBER, Yahoo Finance

Stock market (2)

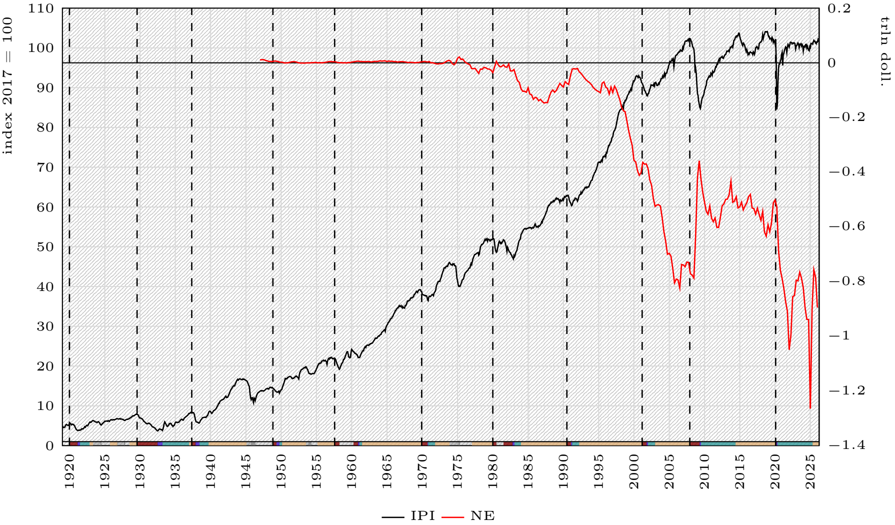
Observation

Stock market indices generally repeat the dynamics of the industrial production index, characterized by an earlier decline at the end of the growth phase since the pricing of financial assets is based on the level of expected income.

Finding

The dynamics of stock market indices reflect the increased role of fictitious capital in the context of expanded capitalist reproduction under the domination of financial capital as well as the process of depreciation of financial capital in the context of the crisis of overproduction.

Net exports (1)



Source: FRED, NBER

Net exports (2)

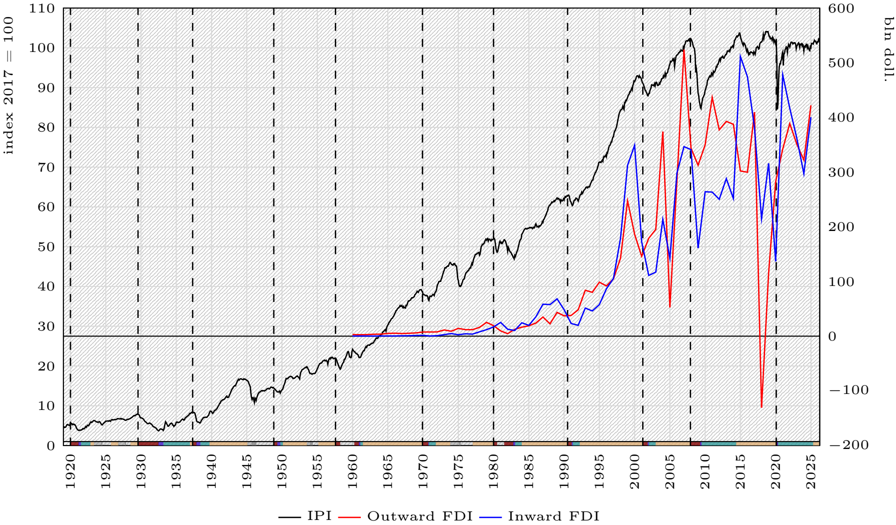
Observation

Net exports in the US economy have a long-term downward trend, but during the crisis phase temporarily recover under the influence of a reduction in national income and import volumes.

Finding

The dynamics of net exports clearly illustrates the compression of export potential, domestic market and economic turnover relative to other countries during the crisis of overproduction.

Foreign direct investment (1)



Source: FRED, NBER, UNCTAD

Foreign direct investment (2)

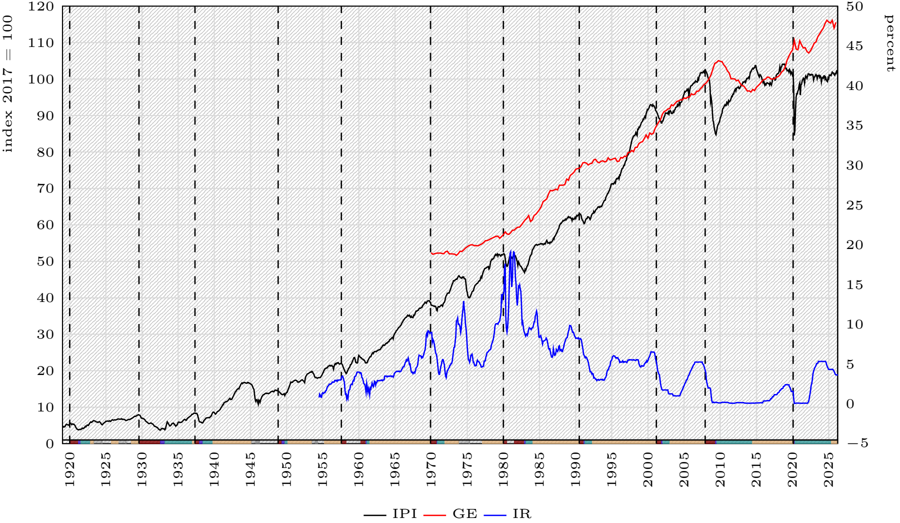
Observation

Outward and inward **foreign direct investment** generally follow the dynamics of private fixed investment: decrease earlier at the end of the growth phase and recover a little later after the recovery phase.

Finding

The dynamics of outward and inward foreign direct investment vividly illustrates international capital flows in relation to the industrial cycle at the national level; a decrease in foreign direct investment at the end of the growth phase can be considered as a leading indicator of the crisis of overproduction.

Government expenditures and interest rate (1)



Source: FRED, NBER

Government expenditures and interest rate (2)

Observation

- **Government expenditures** in the US economy have a long-term upward trend, but during the crisis phase their growth rates increase under the influence of stimulating government policies.
- **Interest rate** increases towards the end of the growth phase to limit inflationary pressures and decreases during the crisis in order to support bank liquidity and credit growth which corresponds to the transition from a restraining to a stimulating monetary policy.

Finding

The dynamics of government expenditures and the central bank's interest rate clearly illustrate the nature of government regulatory policy in the context of the phases of the industrial cycle.

Structure

Results of the research

1. Concept of an economic cycle is defined, approaches to its research and definition are systematized;
2. Key elements of the cycle theory are highlighted, the analysis and systematization of existing theoretical concepts are carried out on their basis;
3. Role of the industrial cycle as a specific form of dialectical development of the capitalist economy is justified;
4. Process of reproduction of fixed capital as a key factor of the industrial cycle is described and modeled;
5. Concrete historical analysis of crises and cycles in the US economy has been carried out;
6. Empirical analysis of the dynamics of indicators and cycle measures in the US economy has been performed.